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Gas Exchange and Control of Breathing in the Electric Eel, *Electrophorus electricus**

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Summary. 1. The electric eel *Electrophorus electricus* is an obligate air breather. Its mouth is structurally adapted for air breathing by an extensively diverticulated and richly vascularized oral mucosa. Air is regularly taken into the mouth and later expelled at the opercular openings. The present investigation concerns the respiratory properties of blood, the dynamics of gas exchange and the control of breathing in the electric eel.

2. Fishes were anesthetized and catheters implanted for sampling of gas in the mouth and blood from the jugular vein draining the mouth respiratory organ, and from a systemic artery. A blood velocity transducer was implanted on the ventral aorta. Following recovery, gas from the mouth, blood gases, blood pH as well as other respiratory and circulatory parameters, were monitored during normal breathing cycles and in response to low and high oxygen tensions in both the aquatic and aerial environment surrounding the fish. In addition, the fish were exposed to a CO_2 enriched environment.

3. Table 2 summarizes the respiratory properties of blood. The high oxygen capacity and oxygen affinity may be an adaptive measure against the mixed conditions of arterial blood. The oxygen capacity was largely unaffected by CO_2.

4. *Electrophorus* showed arterial CO_2 tensions higher than for typical aquatic breathers and other air breathing fishes studied. PCO_2 is increased due to the shunting of blood from the mouth organ to the venous side of the systemic circulation. For the same reason arterial oxygen tensions are normally much below the P100 value. The blood bicarbonate concentration is higher than in typical aquatic breathers.

5. The gas exchange ratio was very low for the mouth respiratory organ and tended to decrease still further in the intervals between air breaths. The gills and/or skin are hence important for CO_2 elimination.

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6. The interval between air breaths rarely exceeded two minutes in intact free-swimming fish surrounded by aerated water and normal ambient air. The fish was irresponsive to changes in oxygen and CO_2 tensions in the water, but breathing of hypoxic and hypercarbic atmospheres caused marked and very prompt increase in the rate of air breathing. Inhalation of a hyperoxic atmosphere caused a depression of air breathing.

7. Heart rate and cardiac output values were higher than earlier reported values for fish. Calculations showed that marked changes occured in the fractional distribution of the cardiac output related to the phase of the breathing cycle and the oxygen tension in the mouth organ.

8. When long intervals prevailed between air breaths the heart rate and cardiac output declined late in the breath interval. Inflation of the mouth organ with oxygen or nitrogen both prompted cardioacceleration and increased blood flow. The changes were of reflex nature and caused by pressure or tension changes inside the mouth.

A variety of tropical fresh water fishes show adaptations toward direct utilization of atmospheric oxygen. In a majority of these, air breathing is an adjunct to aquatic respiration. In others, however, air breathing is obligatory, and these fishes die from asphyxia if prevented from access to air.

The electric eel, *Electrophorus electricus*, lives in muddy rivers and creeks of tropical South America which often become severely oxygen deficient. The fish is an air breather and succumbs if kept from breathing air. The structural adaptations which permit air breathing in *Electrophorus* include an extensively diverticulated and profusely vascularized oral and pharyngeal mucosa. Air is regularly taken into the mouth or buccal cavity by snapping or gulping movements. Expired air escapes from the opercular openings or the mouth before renewal of the air supply.

The gills of the electric eel are markedly degenerated with few and extremely coarse filaments. Aquatic respiration as indicated by branchial movements is normally not observed in adult fish.

The present study concerns the dynamics of gas exchange and the control of breathing in the electric eel.

Anatomy of the Air Breathing Structures in Electrophorus

The structural arrangement and pattern of blood circulation through the air breathing organ in *Electrophorus* have been described in detail earlier (HUNTER, 1861; EVANS, 1929; BÖKER, 1933; CARTER, 1935; RICHTER, 1935) and in the present context only a few features need be emphasized.

The respiratory organ of the mouth is represented by a richly vascularized epithelium, which by papillated projections and foldings has a considerably expanded surface. The vascular papillae are distributed over both the floor and the roof of the mouth. In addition there are smaller prominences present on the branchial arches and portions of the lateral branchial walls. In the floor of the mouth a central raised portion carries three rows of anteroposteriorly arranged papillae. The roof of the mouth has four distinctly raised rows of papillae arranged to fit into the hollows between the papillae of the floor. Thus, when the mouth is closed, the systems of papillae fit into each other forming a labyrinth of passages. In addition to the expanded surface, this structural arrangement ensures short diffusion distances from air in the mouth to the gas exchange surfaces.

As part of the present investigation attempts were made to measure the surface area of the mouth respiratory organ. The heads were separated from the bodies and fixed in 10% formalin. In preparation for measurements, the lower jaw was separated from the upper jaw and the outline of each respiratory surface was drawn in a camera lucida to determine the projected area of the respiratory epithelium. Each part was then imbedded in gelatin and cut in thick sections in a plane normal to the longitudinal axis of the fish. The area of the highly convoluted surface was determined by measuring for each slice the total length of the convoluted outline of the respiratory epithelium and multiplying this by the thickness of the section. After obtaining the surface area for each section, all these were added together to obtain total surface. Table 1 summarizes the results obtained. No corrections have been made for the shrinking effect of formalin on the tissue.

Table 1. *Respiratory surface area*

Specimen	Body weight grams	Estimated body surface[a] (cm^2)	Upper jaw (cm^2)	Lower jaw (cm^2)	Total (cm^2)	Respiratory surface in % of body surface area
A	3200	2170	172.7	147.7	320.4	14.7
B	5000	2920	141.4	230.6	372.0	12.7
C	438	576	47.38	30.07	77.45	13.5
D	360	506	45.26	33.4	78.66	15.5
E	315	463	39.08	25.61	64.69	14.0

[a] Estimated from the expression SA (cm^2): $10 \times$ BW $(\text{gram})^{2/3}$.

Histological examination of the papillae revealed a very rich vascularization. Support for the papillated structures comes from a cartilaginous core and fibrous connective tissue forming the base of the vascularized epithelium. Fig. 1 A shows a schematic drawing of the mouth organ and its blood supply. The inflow to the respiratory surfaces is derived from the branchial arteries which arise in three pairs from the ventral aorta. The most posterior vessel bifurcates to supply the two posterior branchial arches. There are thus four afferent branchial vessels, all of which send branches to the ventral and lateral portions of the mouth organ. The roof of the mouth is supplied by vessels originating from the two first branchial arches on the dorsal side of the gills before these join the other branchial vessels to form the dorsal aorta.

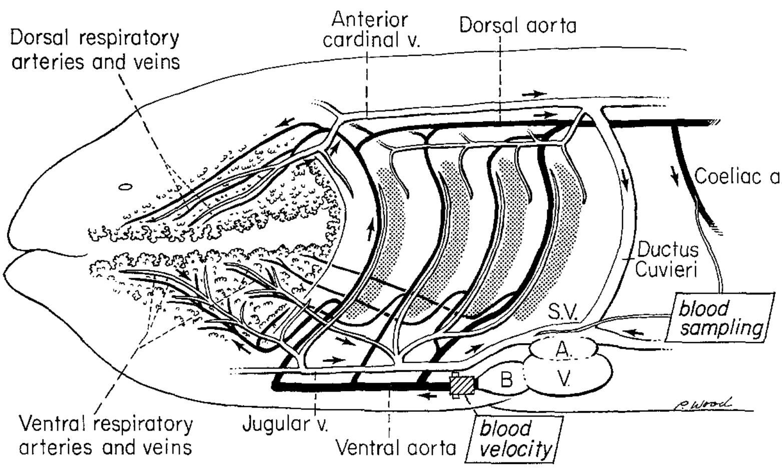

Fig. 1A. Schematic drawing of the mouth respiratory organ and its connection with the heart and central circulation in the electric eel. The arrows indicate the direction of blood flow. The placements of catheters for blood sampling and the blood velocity transducer are also shown

The primary branchial apparatus is marked by a striking reduction in the size of the gill filaments. The secondary lamellae are narrow and thick and the coarse respiratory epithelium suggests that the gills are of little or no consequence in gas exchange. This is also borne out by the direct passage of the branchial vessels through the gill region without the usual resolution into branchial exchange vessels. The striking reduction in surface area and the great thickness of the respiratory epithelium in the gills are common characteristics of fishes which have efficient accessory air breathing organs.

While the mouth respiratory organ in *Electrophorus* receives its afferent blood supply directly from the heart, like the gills in fishes in general, the efferent circulation differs fundamentally from the general piscine pattern. Rather than joining the branchial vessels on their dorsal side in making up the dorsal aorta perfusing the general systemic circulation, the drainage from the mouth organ is connected to the general anterior venous inflow system to the heart. Thus blood from the floor and lateral parts of the mouth organ drains into the jugular veins, whereas the dorsal parts of the mouth organ are connected to the anterior cardinal veins. Both the jugular and anterior cardinal veins, however, receive contributions from systemic veins and the oxygenated blood coming from the mouth organ will be mixed with general systemic venous blood. In the sinus venosus additional venous admixture will occur with blood from the posterior cardinals and the hepatic veins. The blood leaving the heart is thus mixed blood which will undergo no or negligible further gas exchange in the atrophied gills before it reaches the dorsal aorta for systemic arterial distribution.

Materials and Experimental Procedures

Five large (4—10 kg) electric eels, *Electrophorus electricus*, were caught on hook and were kept for several days after capture before experimentation was started. They appeared in excellent health, save one specimen that had a tail wound inflicted by one the other eels.

In preparation for cannulations and implantations the fishes were anesthetized by submersion in MS222. Following anesthesia, the fish was incised mid-ventrally just anterior to the anus. The coeliac artery was dissected free and cannulated with polyethylene tubing (PE100) in a non-obstructive manner so that continued flow in the vessel was allowed. The ventral aorta for implantation of a blood velocity transducer and the jugular vein for catheterization were reached through another incision medio-lateral to the ventral midline in the region of the heart. Catheterization of the jugular vein proved difficult due to the dorsal course of this vessel at its confluence with the sinus venosus. Again the catheter (PE100) was passed through the vessel wall in upstream direction without interfering with the passage of the blood. The blood velocity transducer (8 mm diameter) fitted snugly around the ventral aorta. No constriction of the cardiac outflow tract nor interference with the pericardial wall was apparent. The catheters and leads from the blood velocity transducer were guided out through the incisions, which were closed carefully. Gas sampling from the mouth was arranged by placing a loop of large bore polyethylene tubing (PE 240) in the oral chamber. All cannulations and the blood velocity transducer implants allowed sampling and recording for several days. The placement of the catheters and the blood velocity transducer are indicated in Fig. 1A.

The fishes were allowed to recover for several hours after the operations and experimentation was continued for several days. In addition to offering access to sampling, the catheters allowed injections of drugs and recordings of blood pressure and heart rate.

Blood gases, blood pH, and circulatory and respiratory parameters were monitored during normal breathing cycles and in response to changes in the aquatic and aerial environment. The changes included deoxygenation and hyperoxygenation of the two environments (water and air) as well as exposure to a CO_2 rich (5% CO_2 in air) atmosphere or CO_2 bubbled through the water.

The investigation also included an analysis of the respiratory properties of the blood. In one case blood was obtained from a fish serving solely as blood donor. Additional blood was obtained from the specimens which served as experimental animals.

Methods

Partial pressures of O_2 and CO_2 were measured with a Beckman 160 gas analyzer using the oxygen macro electrode and the Severinghaus CO_2 electrode, both mounted in microcuvettes. The PO_2 electrode was calibrated with tonometered blood samples. The PCO_2 electrode was calibrated with known gas mixtures. Blood pH was measured with a Beckman micro assembly. All gas analysis and pH measurements were made at the temperature prevailing in the experimental tank.

Heart rate and blood pressure were measured using Statham pressure transducers.

Blood velocity was measured with a Doppler shift ultrasonic blood velocity meter (FRANKLIN et al., 1964, 1966). Its application in fishes has been described by JOHANSEN et al. (1966). The system was used for telemetry with the Doppler

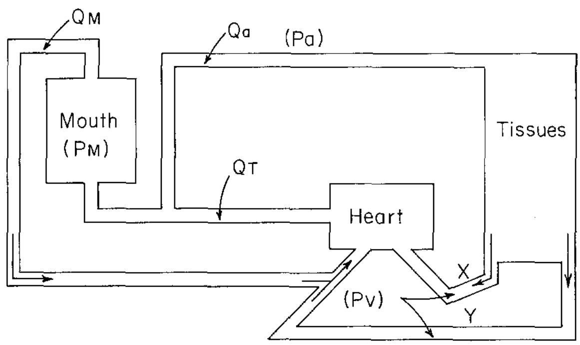

Fig. 1 B. A diagrammatic outline of the circulation including the symbols used for the calculations of relative blood flow changes

frequency shift modulating a 100 Mc/s FM transmitter. The signal was demodulated by an FM receiver and recorded on an Offner Dynograph after conversion to an analog voltage by a frequency to voltage converter (Hewlett-Packard Co.). Blood velocity was computed after electric calibration of frequency shifts or blood flow was calibrated by injecting measured amounts of blood past the transducer *in situ* after the fishes were sacrificed at the termination of the experiments.

Relative changes in blood flow distribution could be calculated from the changes in blood gas composition of the samples drawn from the various blood vessels. These calculations were based on certain assumptions, the validity of which influence the derived values. However, the general trends in relative blood flow distribution are not affected. The symbols used are shown in the Fig. 1 B. The following assumptions were made:

1. The blood leaving the heart is fully mixed with no preferential channeling of blood through the heart.

2. Blood returning to the heart in different systemic veins has the same O_2 content (before being admixed with blood from the respiratory organ of the mouth) (X and Y blood equal).

3. Due to lack of samples of systemic venous blood (X and Y blood), a constant A—V difference of 5 vol.-% is assumed.

4. We assume equilibrium of O_2 tensions of the blood leaving the mouth and the air in the mouth ($P_{M_{O_2}}$).

When total blood flow from the heart equals the sum of flow to the mouth and flow to the systemic arteries (Fig. 1 B) ($Q_T = Q_M + Q_a$) and when systemic arterial blood flow equals all systemic venous return ($Q_a = Q_x + Q_y$) the following derivations can be made:

$$Q_T S_a = Q_M S_M + Q_T S_v - Q_M S_v$$

$$Q_T (S_a - S_v) = Q_M (S_M - S_v)$$

$$\frac{Q_M}{Q_T} = \frac{S_a - S_v}{S_M - S_v}$$

$$\frac{Q_M}{Q_T} + \frac{Q_a}{Q_T} = 1$$

$$\frac{Q_a}{Q_T} = \frac{S_M - S_a}{S_M - S_v}.$$

Symbols: S_a: % Oxygen saturation in arterial blood. S_M: Oxygen saturation in blood leaving mouth. S_v: Oxygen saturation in systemic venous blood.

Hemoglobin content was measured by spectrophotometry. The respiratory properties of blood were established according to methods described earlier (Lenfant and Johansen, 1965).

Results

Respiratory Properties of the Blood

Blood characteristics of the five specimens used are shown in Table 2. Animal 1 should be considered the most representative in expressing normal values, since this specimen was used as a blood donor only, while blood from the other animals was obtained at different times during experimentation. The differences in respect to hemoglobin content, hematocrit, O_2 capacity, and standard bicarbonate, presumably reflect the different histories of the animals.

Table 2. *Blood characteristics*

Specimen number	Hematocrit[a] (%)	Hemoglobin[a] (gram %)	MCHC[b] (%)	O_2 Capacity (vol.-%)	Standard bicarbonates mM/l pH = 7.55
1	37.5	11.2	29.8	13.6	33.5
2	41.0	11.2	27.4	13.9	12.5
3	29.0	7.9	27.2	9.6	23.7
4	29.0	8.1	28.0	—	27.2
5	25.0	6.2	25.0	—	—

[a] Measurements made by Dr. D. W. Allen.
[b] Mean corpuscular hemoglobin concentration.

Fig. 2 shows an O_2-*Hb* dissociation curve based on average values from two specimens. The insert to the figure shows the magnitude of the Bohr effect.

The pH-bicarbonate relationship was determined for each animal and is pictured in Fig. 3. The large spread among the four animals is probably attributable to their different condition. It should be noted that

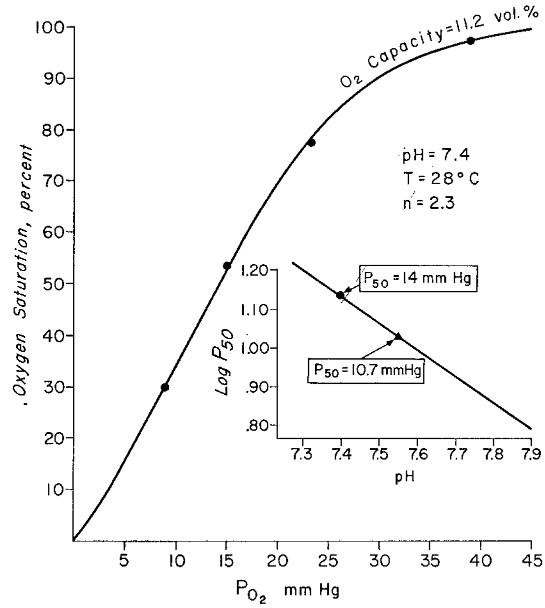

Fig. 2. Oxy-hemoglobin dissociation curve of the electric eel. The curve is based on average values in blood from two specimens. The insert illustrates the Bohr effect; the triangle shows the P 50 at the normal blood pH of the electric eel. For comparison the dot indicates the P 50 value at pH 7.4

the actual buffering capacity expressed as the slopes of the four best fitted lines is fairly similar in spite of the differences in bicarbonate concentration.

Pattern of Breathing

Breathing in *Electrophorus* consists in periodic ascents to the surface where air is snapped or gulped into the mouth. The fish never remains at the surface but sinks back to the bottom. Excess gas or expired gas is released at intervals from the posterior margin of the opercula.

The intervals between ascents to the surface for intake of air showed a great variability, yet a very steady rhythm could prevail for long periods. If prevented from reaching the surface, the fish would struggle violently. Prolonged prevention from air breathing is known to kill the fish. Table 3 shows the mean interval between breaths and that it becomes longer with the time elapsed after surgery.

The breathing pattern turned out to be very sensitive to changes in the ambient gas composition. Experiments to show this were arranged in three ways. Firstly, the ambient water was altered by bubbling nitrogen or 6.5% CO_2 in air or pure oxygen respectively through the aquarium. These procedures evoked no changes in the breathing pattern. Secondly,

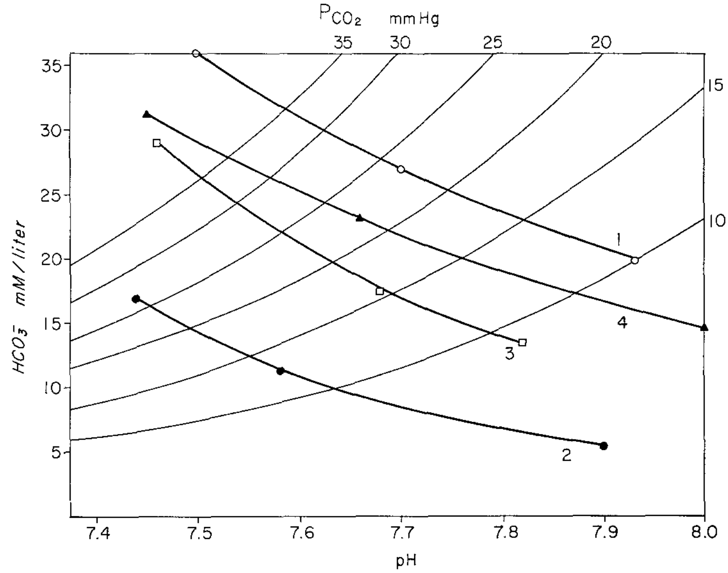

Fig. 3. Bicarbonate-pH relationship in electric eel blood from 4 specimens. PCO_2 isopleths are shown. Note that the bicarbonate-pH slopes which represent the buffering capacity are similar. However, the bicarbonate concentration at any given pH is different for each specimen depending on the condition of the animal. Blood from number 1 was taken while the fish was intact and unanesthetized. Blood from number 2 was collected during anesthesia in connection with the surgical implantation of catheters. Blood samples from fish 3 and 4 were collected two days after surgery

the gas composition in the oral chamber was altered with the fish in underwater position by flushing the respective gas mixtures at constant rate through the mouth catheter, letting the excess gas escape via the mouth or opercular openings. Thirdly, the animals were observed and breathing rates, blood velocity, and heart rates recorded in conjunction with spontaneous surfacings into hypoxic, hyperoxic, or CO_2 enriched atmospheres.

Fig. 4 shows the changes in frequency of breathing when the fish was breathing various O_2 concentrations. These fishes were undisturbed, and the change in breathing frequency was entirely voluntary. It is obvious that the fish is acutely responsive to hypoxic stimulation and that hyperoxic breathing prolongs the intervals between breaths. Fig. 5 shows a continuous tracing of ventral aortic blood velocity during hypoxic breathing and after transition to breathing normal air. Each air breath is apparent as a larger excursion on the tracing. Following the first breath of air, the interval between air breaths is immediately more than doubled. Fig. 6 shows similar tracings illustrating the change in breathing

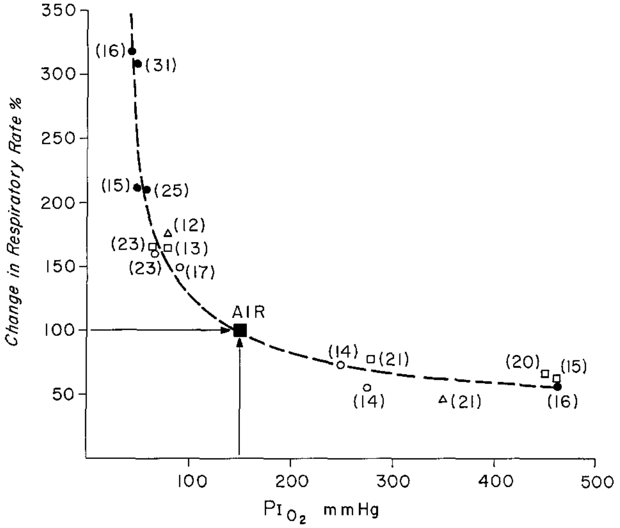

Fig. 4 Relationship between the average frequency of air breathing expressed in per cent of the rate when breathing normal air, and the partial pressure of oxygen in inspired air. The parenthetical values express the number of consecutive air breaths used to calculate the average

Table 3. *Interval between air breaths*

Animal number	Day	Time	Observation	Number of breaths	Mean interval (seconds)	Standard deviation
II	1st	AM	After surgery	34	56.8	68
	2nd	AM		24	87.7	29
		AM		24	95.4	30
		PM		27	71.8	19
	3rd	AM		48	67.6	29
III	1st	AM	After surgery	43	94.9	36
	2nd	AM		27	29.9	6
		AM		20	41.5	10
		PM		26	62.5	11
		PM		37	63.3	24
	3rd	AM		22	97.5	35
		PM		30	103.8	40
IV	1st	AM	After surgery	33	40.1	7
		PM		25	48.4	14
		PM		32	68.0	12
	2nd	AM		21	87.5	31
		PM		28	79.3	15
		PM		33	78.3	32
V	1st	PM	Surgery AM	44	128.8	39

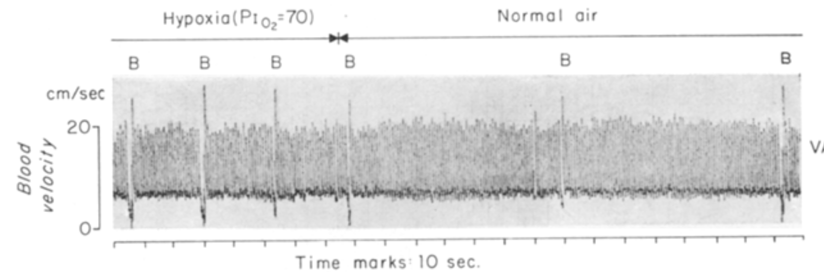

Fig. 5

Fig. 5. Continuous tracing of blood velocity in the ventral aorta of the electric eel. The larger excursions occur with each breath of air. Note the prompt change in the rate of air breathing following the first breath of normal air after a period of hypoxic breathing

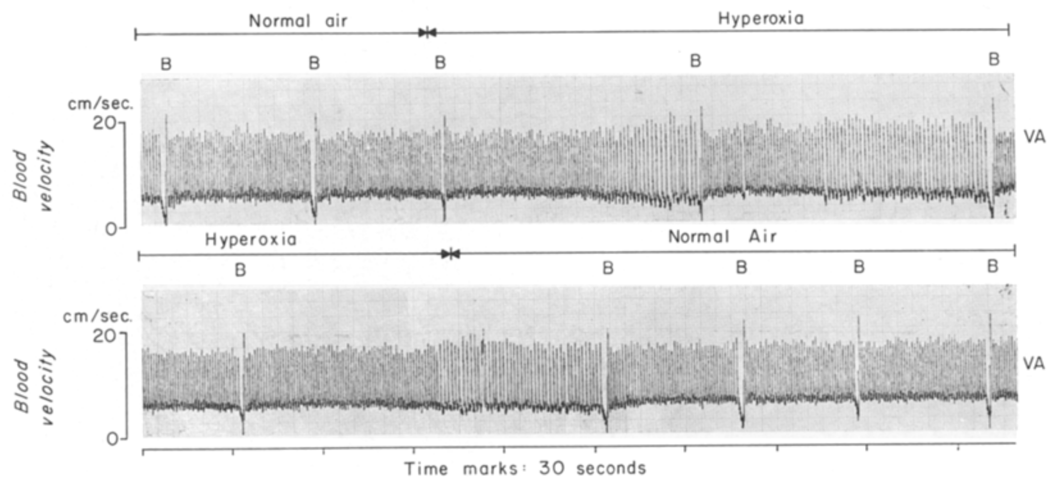

Fig. 6

Fig. 6. Continuous tracing of ventral aortic blood velocity. The larger excursions occur with each breath of air. Note the prompt increase of intervals between air breaths following the first breath of an oxygen enriched atmosphere

pattern when shifting from air to an hyperoxic atmosphere. Again, the change in breathing is established after one single breath of the new atmosphere.

Breathing a CO_2 rich atmosphere (6.5% CO_2 in air) exerts a pronounced stimulatory effect on the rate of air breathing (Fig. 7). Note, however, that the effect is much smaller than the response to hypoxic breathing.

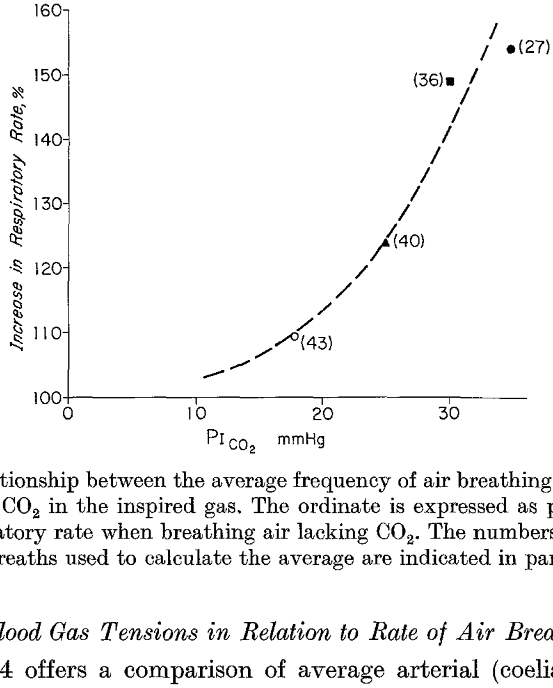

Fig. 7. Relationship between the average frequency of air breathing and the partial pressure of CO_2 in the inspired gas. The ordinate is expressed as per cent change from respiratory rate when breathing air lacking CO_2. The numbers of consecutive air-breaths used to calculate the average are indicated in parentheses

Blood Gas Tensions in Relation to Rate of Air Breathing

Table 4 offers a comparison of average arterial (coeliac) blood gas tensions and pH during normal air breathing, and during breathing of

Table 4. *Mean arterial blood gas*

Animal number	Inspired gas											
	air						High PO_2					
	PO_2[a]		PCO_2[a]		pH		PO_2		PCO_2		pH	
	n	$\bar{m}$	n	$\bar{m}$	n	$\bar{m}$	n	$\bar{m}$	n	$\bar{m}$	n	$\bar{m}$
II	27	23	—	—	17	7.51	7	36	—	—	6	7.45
III	13	21	11	26.4	5	7.55	9	24	5	30.5	—	—
IV	15	20	8	28.5	7	7.60	6	25	4	28.9	4	7.57
V	15	20	15	28.1	—	—	15	28	15	36.8		
Mean		21		27.7		7.55		28.2		32.1		7.51

[a] All PO_2 and PCO_2 in mm Hg.

hypoxic, hyperoxic, and hypercarbic atmospheres. The mean arterial PO_2 during air breathing is relatively low, averaging 21 mm Hg. During breathing of an hyperoxic atmosphere (PO_2 150 to 450) the arterial PO_2 increased (average 28.2 mm Hg). Associated with this increase was a rise in arterial PCO_2 and a drop in pH, most likely caused by the decreased ventilation following hyperoxic breathing. When a hypoxic gas mixture is inspired, a marked decrease occured in the arterial O_2 tension (average of 12.6 mm Hg). This decrease falls within the steepest portion of the O_2-*Hb* dissociation curve and represents a drastic reduction in arterial O_2 saturation. The reduced arterial PCO_2 and the increase in pH during hypoxic breathing can be ascribed to the increased frequency of air breathing. Breathing a CO_2 rich atmosphere results in a slight increase in arterial PO_2 and a surprisingly small elevation of arterial PCO_2. The sampling of blood underlying Table 4 was done randomly within the breathing intervals and this may account for the large variability. Fig. 8A and B, allow an evaluation of the variability to be expected from such random sampling. Fig. 8A shows a clear grouping of the arterial tensions relative to the various gas mixtures used. Fig. 8B, expressing the arterial PCO_2 changes within breathing cycles corresponding to the three types of external breathing conditions, shows a much larger variability and overlap, and a grouping of the points is barely noticeable.

Changes in Mouth Gas Composition and Blood Gases with Time

Fig. 9 is a composite plot of the time course of the PO_2 and PCO_2 inside the mouth following several breaths of ambient air in four individuals. While the O_2 tensions show a nearly linear decrease, the corresponding rise in CO_2 tensions is steepest in the early phase and changes to a slower rise after about 30—40 sec.

tensions (mm Hg) and pH

Inspired gas											
Low PO_2						High CO_2					
PO_2		PCO_2		pH		PO_2		PCO_2		pH	
n	$\overline{m}$	n	$\overline{m}$	n	$\overline{m}$	n	$\overline{m}$	n	$\overline{m}$	n	$\overline{m}$
7	12.1	—	—	7	7.58	9	28	—	—	9	7.32
13	13.1	13	20.5	—	—	12	19	10	33	4	7.50
11	11.6	5	20.0	5	7.71	11	21	11	33.7	9	7.53
14	14.1	14	26								
	12.6		22.2		7.65		22.7		33.3		7.45

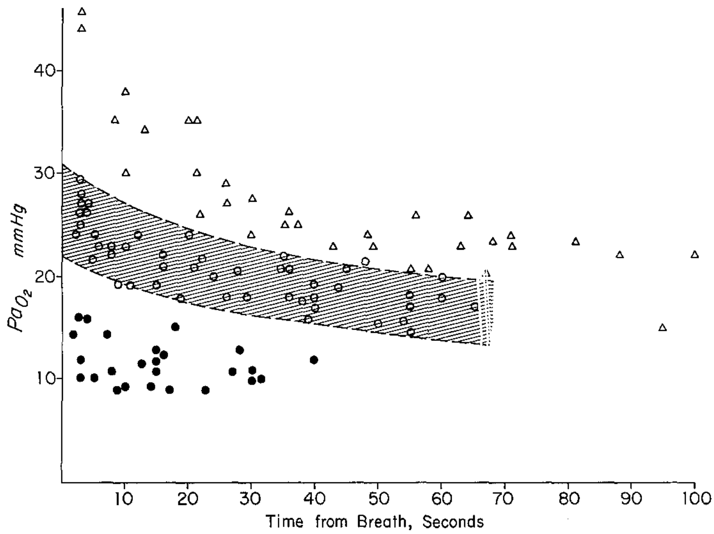

Fig. 8A. A composite plot of the decrease in arterial oxygen tension during intervals between consecutive air breaths. The shaded area corresponds to values obtained when the animals were breathing air. The values marked as triangles were obtained during hyperoxic breathing ($P_IO_2 > 350$ mm Hg) and the dots when the animals were breathing a hypoxic gas mixture ($P_IO_2 < 60$ mm Hg).

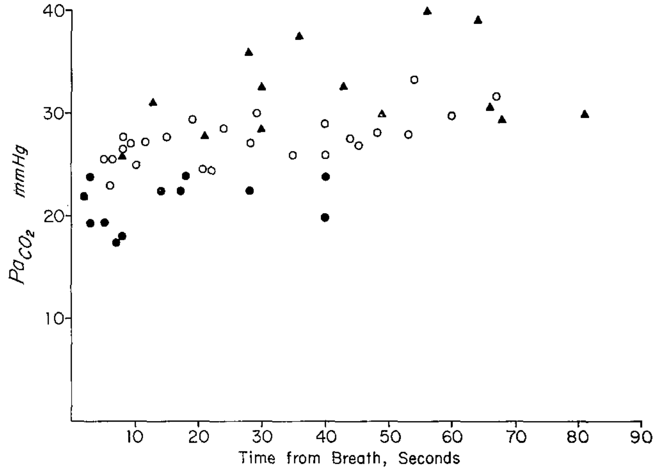

Fig. 8B. A composite plot of the increase in arterial CO_2 tension during intervals between consecutive breaths. Open circles express values obtained during breathing air, triangles correspond to breathing hyperoxic and dots to breathing a hypoxic gas mixture

Changes of gas composition in the mouth, and of gas tensions in systemic arterial blood and jugular venous blood (the latter represents part of the venous drainage from the respiratory mouth organ [Fig. 1]) are illustrated for breathing of ambient air or a hypoxic atmosphere or a hyperoxic atmosphere in Figs. 10, 11 and 12. Additional computed information is listed at the top of each graph with arrows marking the time of correspondence with the measured information. The computed information includes the gas exchange ratio, the relative ratio of mouth

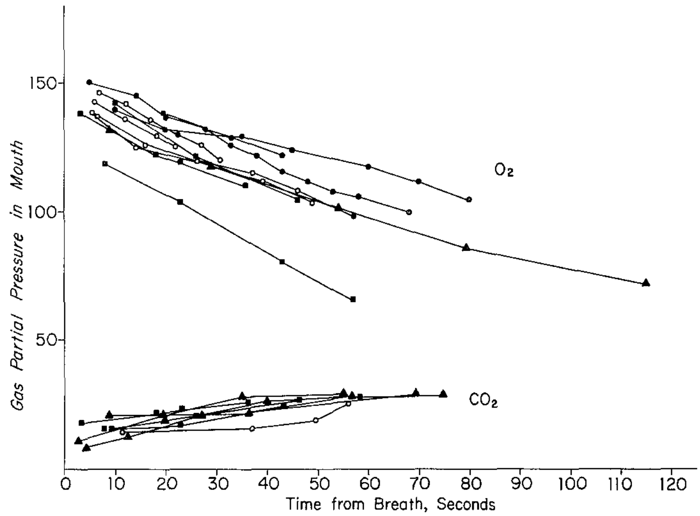

Fig. 9. Changes of oxygen and CO_2 tensions of gas in the mouth as a function of the time elapsed between consecutive breaths in four individuals identified respectively by symbols of squares, triangles and closed and open circles

blood flow (Q_M) to total flow (Q_T) and the ratio of systemic arterial flow (Q_a) to total flow (Q_T). Fig. 10 shows the above parameters during normal airbreathing. During the three minute intervals between 2 breaths, the PO_2 in the mouth (P_MO_2) drops fairly linearly from about 140 mm Hg to about 60 mm Hg. In the same interval, the mouth CO_2 tension (P_MCO_2) rises from about 7 Hg to about 30 mm Hg, with most of the increase taking place in the first 90 sec. Accordingly, there is a striking drop in the gas exchange ratio during the breathing interval from 0.85 ten sec. after the breath to 0.20 two and one half minutes later. The blood gas values suggest that most of the blood leaving the heart goes to the mouth from where it is shunted back to the heart. Importantly, the ratio of mouth blood flow to total flow changes from about 75% right after the breath to about 60% at the end of the breath interval. The fraction of the total flow being diverted past the mouth to the dorsal aorta varies inversely (Fig. 10) and changes from 25% to about 40%.

Fig. 10 reveals a huge difference between the O_2 tension in the mouth and that in the jugular vein entering the heart. The steep gradient could result from a diffusion barrier from gas to blood in the mouth, but the shape of the dissociation curve is probably more decisive. The PO_2 of

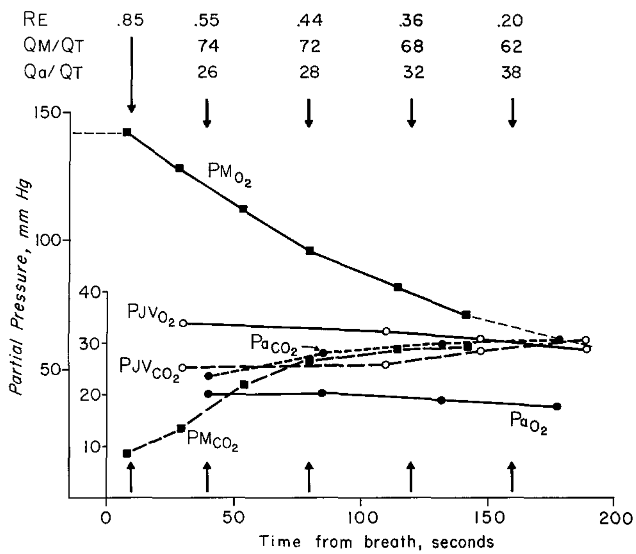

Fig. 10. Changes in O_2 and CO_2 tensions of the mouth gas, (M) systemic arterial blood (a) and blood from the jugular vein (jv) between two consecutive air breaths. Listed on top of the figure are, computed values of the gas exchange ratio (RE), the fraction of the total cardiac output going to the mouth (Q_M/Q_T) and to the systemic arteries (Q_a/Q_T). The vertical arrows indicate the gas partial pressures used for these calculations

blood draining the mouth is high and is therefore located on the flat portion of the O_2-Hb dissociation curve. Thus, when this blood mixes with the O_2 depleted systemic venous blood in the jugular vein, PO_2 must decrease greatly. When PO_2 in the mouth decreases, the difference in PO_2 between the jugular venous blood and the systemic arterial blood becomes smaller because the values are located on the steep part of the O_2-Hb dissociation curve and the difference in O_2 tension becomes less apparent.

Fig. 11 is similar to Fig. 10 but is based on data obtained during hyperoxic breathing. The general pattern is similar, but due to a relatively moderate increase of the O_2 tension in the jugular vein, the difference between the jugular vein and the systemic arterial blood is higher. The rate of decrease in mouth PO_2 is more rapid than during air breathing.

This must in part be related to an increase in the total blood flow going to the mouth organ. Values at the top of Fig. 11 testify to this, as the mouth flow represents 73% of total flow during hyperoxic breathing against 62% during air breathing, in both cases compared at the same time (160 sec) after a breath.

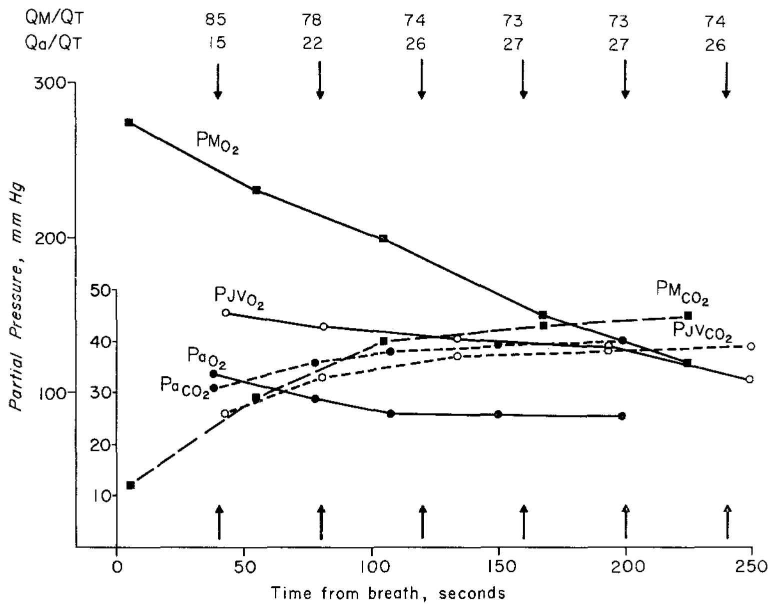

Fig. 11. Changes in O_2 and CO_2 tensions of mouth gas (M) systemic arterial blood (a) and blood from the jugular vein (jv) between consecutive breaths of a hyperoxic atmosphere. Listed on top of the figure are calculated values of the fraction of total blood flow going to the mouth (Q_M/Q_T) and to the systemic arteries (Q_a/Q_T)

When the fish breathes a hypoxic gas mixture (Fig. 12), the general pattern is the same. However, now the rate of change in P_MO_2 is markedly reduced, because of the relative changes in the flow distribution to the mouth. This time a much smaller fraction of the total blood flow is diverted to the mouth. The breath intervals are much shortened as described earlier, and already after 75 sec or about half the length of an average breath interval during breathing of air, the fraction of mouth flow to total flow is down to about 52%. A corresponding figure during air breathing would be 72%. The PO_2 gradient between jugular venous blood and systemic arterial blood is now considerably reduced.

Fig. 13A and B illustrate further the relationship between the relative blood flow to the mouth and PO_2 in the mouth and the phase of the breathing cycle. Fig. 13A shows that following intake of normal air into the mouth, the fraction of mouth flow to total flow remains relatively unchanged until P_MO_2 has dropped below 100 mm Hg. From then on the flow fraction to the mouth declines. Intake of N_2 into the mouth causes a more prompt reduction in the flow fraction to the mouth organ whereas inhalation of an oxygen rich atmosphere causes an increase in

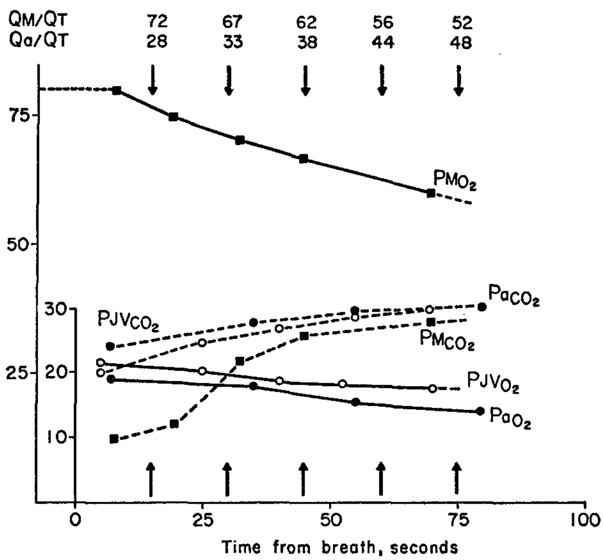

Fig. 12. Changes in O_2 and CO_2 tensions of mouth gas (*M*) systemic arterial blood (*a*) and blood from the jugular vein (*jv*) between consecutive breaths of a hypoxic atmosphere. Listed on top of the figure are calculated values of the fraction of total blood flow going to the mouth (Q_M/Q_T) and to the systematic arteries (Q_a/Q_T)

the mouth flow when the P_MO_2 exceeds about 200 mm Hg. It seems significant that the average breath duration as revealed in Fig. 9 and Table 3 will result in a relatively stable flow distribution between the mouth organ and the systemic circulation. It was documented earlier (Figs. 4 and 7) how breathing of foreign atmospheres (hyperoxic, hypoxic, and hypercarbic) markedly changed the pattern of breathing. Fig. 13A and B provide additional information on how such breathing affects the relative blood flow to the air breathing organ in the mouth.

Cardio-Vascular Responses

The heart rates in *Electrophorus* at ambient temperatures of 28—30° were high and fluctuated between 65 and 75 per min. Corresponding mean arterial blood pressures varied between 30 and 45 cm H_2O in resting fish. Cardiac outputs under resting conditions fluctuated between 40 and 70 ml/kg-min.

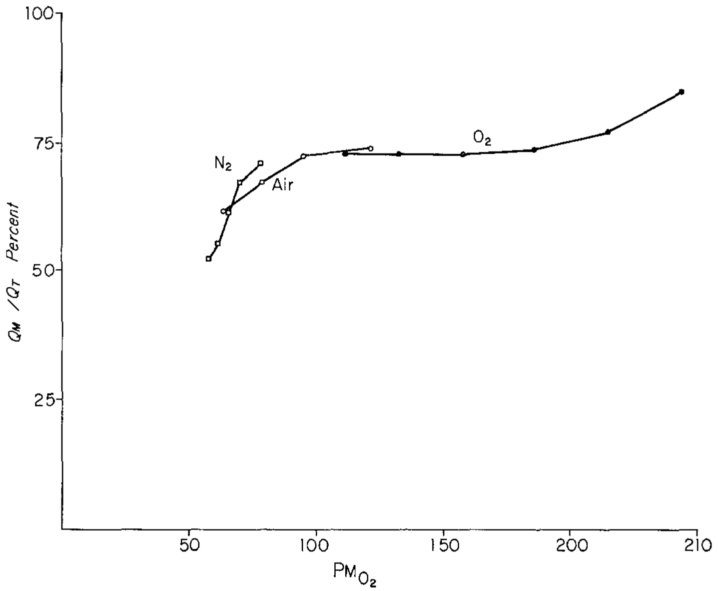

Fig. 13A. Relationship between the fraction of the cardiac output going to the mouth (Q_M/Q_T) and the partial pressure of oxygen in the mouth gas (P_MO_2). The three curves were obtained while the animal was breathing a hyperoxic gas mixture (O_2), ordinary ambient air (Air) or a nitrogen enriched, hypoxic (N_2) gas mixture. The mouth gas and blood samples underlying the calculations were obtained during the interval between consecutive breaths for each of the 3 established curves.

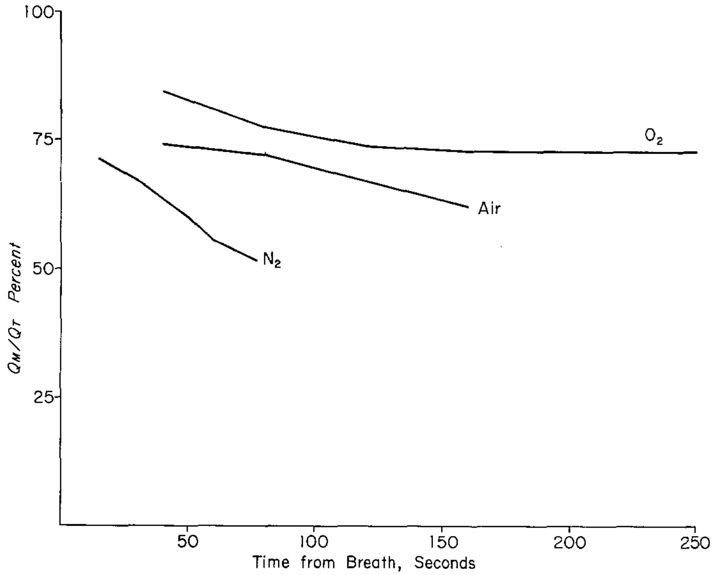

Fig. 13B. Changes in the fraction of the cardiac output going to the mouth (Q_M/Q_T), with time between consecutive breaths. The three curves were obtained while the animal was breathing a hyperoxic gas mixture (O_2), ordinary ambient air (Air) or a nitrogen enriched hypoxic (N_2) gas mixture

Cardio-vascular and respiratory events are known to be strongly interrelated in fish, and this tendency was also apparent in *Electrophorus*. In particular heart rate, blood pressure, and cardiac output showed marked spontaneous changes depending on the phase and duration of

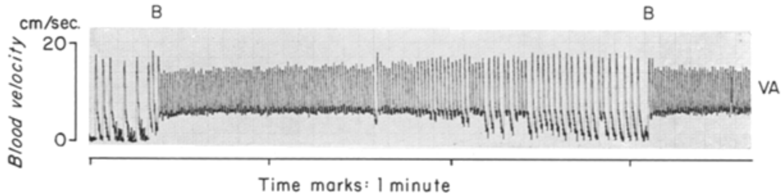

Fig. 14. Ventral aortic blood velocity between two air breaths (*B*). When the interval between breaths is long the heart rate and blood velocity decline toward the end of the breath interval

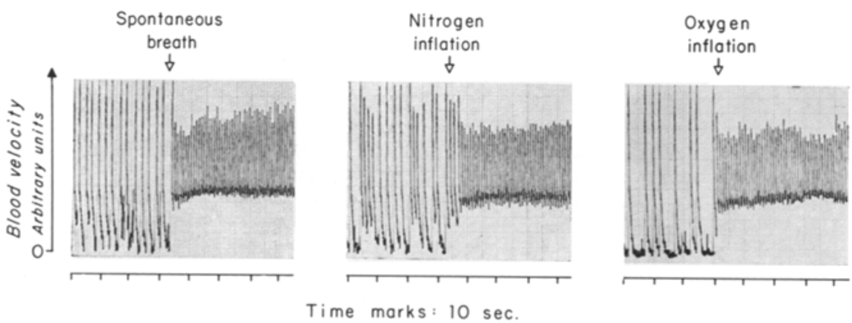

Fig. 15. Changes in heart rate and ventral aortic blood velocity occuring spontaneously and following inflation of the mouth with nitrogen and oxygen. The tracings were obtained during recovery from anesthesia when the intervals between breaths were especially long

the breathing cycle. When the breath intervals were longer than about one minute, both heart rate and blood velocity progressively decreased (Fig. 14). At the next breath, however, the heart rate and blood velocity increased instantaneously and became re-established at the values prevailing shortly after the preceding breath. This cyclic phenomenon occurred as a normal event when the breath to breath intervals were long and also showed up when the duration of the breath interval was prolonged by hyperoxic breathing as shown in Fig. 6. A comparison of cardiac output at the beginning and the end of a long breath interval indicates at least a five-fold decrease. If the prevailing breath to breath intervals were short, no time-dependent blood velocity changes were found.

The sudden change in cardio-vascular performance elicited by an air breath suggests a reflex response. Information as to the nature of this response was gained from animals recovering from anesthesia in which the breath-to-breath intervals were especially long.

Fig. 15 shows the conspicuous change in heart rate and blood velocity elicited by a breath during such conditions. The two possible modes for elicitation of the reflex could be via a mechanical or a chemical type of stimulus. Artificial inflation via the catheter in the mouth caused a similar response when either nitrogen or oxygen was injected (Fig. 15). The response thus appeared to be elicited by the volume or pressure of the gas in the mouth, rather than its chemical composition.

Discussion

The mode of air breathing in the electric eel is different from that in all other air breathing fish. The papillated evaginations and the profuse vascularization of the buccal mucosa are unique features. The delicate nature of a respiratory surface with its susceptibility to mechanical injury makes the location of the air breathing organ in the entire buccal cavity seem very disadvantageous. Perhaps it is the remarkable method of predation by stunning the victim with a powerful electric shock that has permitted the carnivorous electric eel to modify its entire buccal cavity for respiratory purposes. The teeth at the anterior and posterior extremes of the mouth are useful only in holding a stunned victim during the process of swallowing, but biting or chewing on a struggling victim would seem impossible with such delicate vascular structures in the mouth.

Before discussing the physiological implications of our findings, it seems imperative to emphasize the pattern of blood circulation in *Electrophorus*. The efficiency of any respiratory organ depends on the pattern of blood perfusion through the organ. When the respiratory organ is not placed in direct series with the heart and the systemic circulation, the usefulness of external gas exchange will depend on the extent to which blood recirculates to the respiratory and systemic vascular circuits.

Electrophorus in this respect seems to be at a structural disadvantage since the entire volume of blood perfusing the respiratory organ is returned to the heart and mixed with the systemic venous blood before it is redistributed to the systemic vascular beds and the respiratory circuit. The perfusion pattern would appear more logical if the efferent circulation from the respiratory organ would drain directly into the dorsal aorta giving rise to the entire systemic circulation. Such a pattern of circulation exists in those fishes which use aquatic breathing, and also in a few air breathing fishes which are able to use their gills directly or have respiratory branchial diverticula for aerial gas exchange

Symbranchus marmoratus, (JOHANSEN, 1966), *Hypopomus brevirostris*, (CARTER and BEADLE, 1931) and *Clarias* (RAUTHER, 1910).

Most structural adaptations for air breathing in fishes shows a respiratory efferent circulation which is connected to the systemic venous circulation. This suggests that the vascular resistance in organs that serve for aerial gas exchange is so high that most of the propulsive energy imparted to the blood by the heart has been dissipated in the perfusion of the exchange organ. The best solution to this situation is the one that has been favored in the evolution of all terrestrial vertebrates, namely a separate return of the respiratory efferent blood directly to the heart, where septations maintain a separation in the outflow channels and permit the oxygenated blood to be dispatched to the systemic circulation. That such a structural development was favored at a very early stage in the evolution of air breathing is testified to by the extant lungfishes (JOHANSEN et al., 1968; LENFANT and JOHANSEN, 1968).

The hematocrit, hemoglobin content, and oxygen capacity of *Electrophorus* blood are higher than for most teleosts with sluggish habits. WILLMER (1934) reported even higher values for O_2 capacity in *Electrophorus* averaging 19.75 vol.-%. The conclusion that such high values reflect an adaptation towards oxygen deficient ambient conditions (KROGH and LEITCH, 1919) is refuted by the fact that *Electrophorus* is an obligate air breather which uses the atmosphere as the main source of oxygen. It remains an interesting hypothesis that the high oxygen capacity of *Electrophorus* blood is an adaptation to the mixed condition of systemic arterial blood, resulting from the huge shunting of oxygenated blood to the systemic veins. This shunting causes the systemic arteria blood always to remain below full saturation and at times even below half saturation. The usefulness of a high O_2 capacity in such a situation is obvious. The high O_2 affinity of *Electrophorus* blood may similarly represent an adaptive measure against the mixed condition of systemic arterial blood.

Commonly fishes from well oxygenated waters show a marked influence of CO_2 on the hemoglobin affinity for O_2 (Bohr effect) and the O_2 combining power (Root effect) of the blood. It seems a significant adaptation that the O_2 capacity of *Electrophorus* blood is largely unaffected by CO_2. Adaptive changes in the O_2 affinity are less apparent, which is to be expected since a change in the affinity must be a compromise between the usefulness of the Bohr shift in the unloading and the loading of hemoglobin with O_2.

The shift to a greater dependency on air breathing in fishes is in the majority of cases attended by a general reduction in the efficiency of the gills. Due to the lower solubility of oxygen than CO_2 in water, the gill reduction will pose stricter limitations on O_2 absorption than CO_2

elimination in aquatic gas exchange. A shift towards increased air breathing in forms employing a bimodal gas exchange has, however, also been shown to correlate with increased arterial CO_2 tensions (LENFANT, et al., 1966; LENFANT and JOHANSEN, 1967). *Electrophorus* showed high arterial CO_2 tensions, exceeding those in typical aquatic breathers among fishes as well as those of air breathing fishes like the African lungfish (LENFANT and JOHANSEN, 1968). An additional factor causing CO_2 retention in the electric eel is the shunting of blood from the mouth organ directly to the venous side of the systemic circulation.

A general elevation of internal CO_2 tensions poses a requirement for an increased CO_2 combining power and buffering capacity of the blood. Our results indicate that the blood bicarbonate concentration, although variable, is increased compared to conditions in typical aquatic breathers (Table 2, Fig. 3). It merits attention that the buffering capacities, expressed as the slope of the four lines in Fig. 3, are similar in spite of the marked differences in CO_2 combining power. Another study has emphasized the importance of ample recovery time after anesthesia and surgery in the electric eel to rectify the respiratory and metabolic acidosis incurred during such procedures (GAREY and RAHN, personal communication). This may explain the variability in the present data.

The breathing behavior of *Electrophorus* left no doubt that the fish is an obligate air breather. In intact free moving fish after complete recovery from anesthesia the intervals between air breaths in well aerated water rarely exceeded two minutes. BÖKER (1932) reported the frequency of air breathing in uninjured, intact fish as once every 1 to 2 min.

Fishes which employ aquatic breathing in well aerated water, are acutely sensitive to deoxygenated water and greatly augment their respiratory efforts. In contrast, *Electrophorus* was irresponsive to hypoxic conditions in the water. Several factors can be invoked to explain the lack of response in *Electrophorus*. First of all, the method of air intake to the mouth precludes that external water enters the mouth to partake in gas exchange. Thus gas composition in the water cannot influence chemoreceptors located in the mouth or in the blood stream. The possibility that external receptor might be sensitive to O_2 tension in the water, as has been suggested for other fish (SHELFORD and ALLEE, 1913), finds no teleological rationale in the environment of *Electrophorus*, which is more or less permanently hypoxic. A similar response type has been described for the African lungfish, *Protopterus*, another obligate air breather with vestigial gills (JOHANSEN and LENFANT, 1968). However, the Australian lungfish, *Neoceratodus* depending primarily on aquatic gas exchange with gills also for O_2 absorption, responds promptly to deoxygenated water by increased branchial ventilation (JOHANSEN et al., 1967).

When *Electrophorus* surfaced into a hypoxic atmosphere, a marked and immediate increase in air breathing followed (Figs. 4 and 5). Assuming equal volumes of air taken in with each breath, the data indicate a doubling of the ventilation volume when the inspired oxygen tension falls to about 70 mm Hg. There can be little doubt that this response to hypoxic breathing is elicited by stimulation of chemoreceptors.

It is worth noting that inhalation of oxygen rich atmospheres causes a depression of breathing, although much less conspicuous than the stimulation by low oxygen. This finding is important by suggesting the removal of a tonic PO_2 dependent stimulus by the hyperoxic breathing, or expressed differently that normal spontaneous breathing is governed by the changes in oxygen tension of the air in the mouth. This situation shows a correspondence to conditions in higher vertebrates (mammals), where oxygen inhalation is effective in removing the tonic activity of chemoreceptor cells stimulated by the normally prevailing levels of PO_2. In mammals, of course, it has long been established that these chemoreceptor cells are located in the carotid and aortic bodies. These anatomical locations are not well defined in the piscine vascular system, and at present it must even remain uncertain whether the chemoreceptors in *Electrophorus* are located in the blood stream or are sensing directly the quality of the air in the mouth. The rapidity with which PO_2 dependent stimuli influences the breathing pattern is remarkable and favors the idea that receptors in the buccal mucosa, or in the blood at a short distance from the air in the mouth, are involved in the response. Fig. 5 shows that following the very first breath in normal air after hypoxic breathing, the breathing rate is adjusted back to a rate characteristic of breathing normal air. Similarly as shown in Fig. 6 the breathing interval is prolonged with the first breath of hyperoxic air and adjusted back equally promptly at the end of hyperoxic breathing.

Important findings about the CO_2 excretion in *Electrophorus* are revealed in Table 3. When the inspired CO_2 tension was artificially increased (P_ICO_2, 18 to 35 mm Hg), the increase in blood PCO_2 was surprisingly modest and barely exceeded the blood CO_2 levels that resulted when the breathing intervals were prolonged by hyperoxic breathing. These results suggest that the fish possesses avenues for CO_2 elimination directly to the aquatic medium. The low gas exchange ratios (R_E) consistently measured in the mouth offer important support for a bimodal gas exchange in *Electrophorus*.

The tendency for the gas exchange ratio to decrease within a breath to breath interval (Figs. 10—12), expresses that CO_2 elimination to the water may vary, possibly activated by a vasodilatory effect in skin and/or the vestigial gills, by the rising internal PCO_2. It is a general characteristic among fishes which have structural adaptations for air breathing that they retain a functional importance of the gills or utilize the skin for CO_2 elimination. In such bimodal gas exchange, the principal role of the air breathing organ becomes oxygen absorption giving it the very low gas exchange ratio.

The efficiency of air breathing in *Electrophorus* is no doubt severely diminished by the huge vascular shunt from the respiratory organ. Compensatory measures to improve gas transport against this disadvantage may be represented by respiratory properties of the blood such as high O_2 capacity and O_2 affinity. Another compensation may be the high resting values of cardiac output. Earlier values for cardiac output in fishes based upon the Fick Principle or dye dilution methods do not exceed 20–25 ml/kg-min (MURDAUGH et al., 1965; HANSON, 1967). Our values for cardiac output reached as high as 70 ml/kg-min. Differences in temperature and methods of measurement as well as the general scarcity of cardiac output measurements in fishes, should be noted in this comparison.

The calculations based on blood gas values expressing the fractional distribution of cardiac output in *Electrophorus* showed that less than 50% of the output was normally distributed to the systemic arteries. A conceivable distortion of these values due to the assumptions made in the calculations does not create any uncertainty as to the applicability of the relative changes in flow distribution with time. The data thus reveal that the distribution of the cardiac output changes markedly during the period between air breaths. The typical example in Fig. 10 shows that right after a breath the blood flow fraction going to the mouth is at its highest, while later in the breathing interval it declines steadily. Fig. 11 obtained during oxygen breathing displays a similar trend but with an ever higher fraction of the cardiac output being diverted to the mouth organ. Conversely hypoxic breathing (Fig. 12) shifts blood away from the respiratory organ. Thus there are two features that stand out in the distribution pattern of the cardiac output. A high level of mouth oxygen will shift more blood to the respiratory organ. Viewed within the frame of a breath to breath interval, the same tendency is apparent as a reduction of flow to the mouth when the oxygen tension in the mouth decreases. This adjustment has an obvious rationale in promoting the matching process between blood and gas in the mouth. It appears that the ratio of respiratory to systemic blood flow remains relatively unaltered until the oxygen tension in the mouth has dropped to about 100 mm Hg (Fig. 13A). At lower P_MO_2 there is a sharp drop in the fraction of the cardiac output being diverted to the mouth organ. It is significant that a breath rarely occurs while the distribution ratio is unchanged. This is also borne out from the time dependence of the distribution ratio as expressed in Fig. 13B. The direct measurements of blood velocity in the ventral aorta add important information by showing that long breath intervals incur a marked change in the total outflow from the heart (Fig. 14).

The spontaneous changes in the fractional distribution of the cardiac output between the respiratory and the systemic vascular beds must result from vasomotor changes affecting the resistance to flow in the various beds. The relationship of the flow changes to the phase of the breath interval expresses that the control of breathing is integrated with vasomotors reflexes. Earlier studies have indicated that fishes show a close coupling of respiratory and circulatory events (Satchell, 1960; Johansen et al., 1968). In elasmobranchs, pharyngeal mechanoreceptors reflexly influence the cardiac vagal tone (Satchel, 1960). Satchell has suggested that these receptors are important in adjusting the blood and water passage across the gills for optimal efficiency in the counter current exchange process. Recently Johansen et al. (1968) have described marked heart rate and blood flow changes related to spontaneous inflation of the lung in the African lungfish.

The present experiments offer evidence that the heart rate and flow changes associated with the breathing act in *Electrophorus* are of reflex nature. The data shown in Fig. 15 imply that the reflex is elicited by a mechanical stimulus. It is a possibility that the low gas exchange ratio in the mouth normally results in a gradual reduction of intrabuccal pressure, causing the retardation of heart rate and cardiac output during long breath intervals. When the interval is broken by intake or air, intrabuccal pressure is raised which in turn provides the mechanical stimulus that sets off the prompt cardioacceleration and flow increase.

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